

Linear Programming and Learning with Constraints

Chen-Yu Wei

Linear Programming

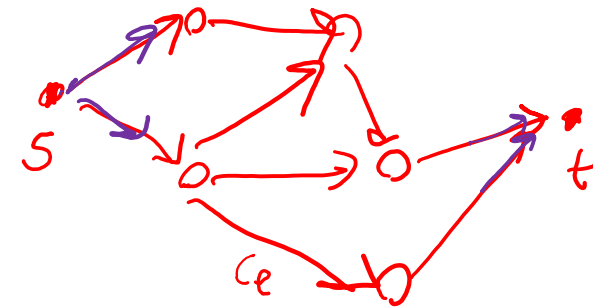
$$\max_{f, F} F$$

subject to

$$\sum_{e \in \text{OUT}(v)} f_e - \sum_{e \in \text{IN}(v)} f_e = \begin{cases} F, & v = s \\ -F, & v = t \\ 0, & v \neq s, t \end{cases}$$

$$0 \leq f_e \leq c_e \quad \forall e \in E$$

$\{f_e\}_{e \in E}$



Linear Programing

$$\max_{f,F} F$$

$$\text{s. t.} \quad Bf = Fb$$

$$0 \leq f \leq c$$

Canonical Form of LP

$$\min_x c^T x$$

$$\text{s.t. } Ax \geq b$$
$$x \geq 0$$

$$\min_x c^T x$$

s.t.

$$a_1^T x \geq b_1$$

$$a_2^T x \geq b_2$$

$\vdots$

$$a_N^T x \geq b_N$$

$$x_1 \geq 0$$

$\vdots$

$$x_d \geq 0$$

$\Rightarrow$

$$Ax \geq b$$

$$x \geq 0$$

Primal and Dual Problems

$$\min_x c^\top x$$

$$\text{s. t. } Ax \geq b$$
$$x \geq 0$$

Primal

$$\max_y b^\top y$$

$$\text{s. t. } A^\top y \leq c$$
$$y \geq 0$$

Dual

Primal and Dual Problems

Primal

$$\alpha = \min_{x \in \mathbb{R}^d} C^T x$$

$$\begin{aligned} \text{s.t. } & Ax \geq b \quad (m \text{ constraints}) \\ & x \geq 0 \quad (d \text{ constraints}) \end{aligned}$$

$$x \in \mathbb{R}^d \quad (d = \text{dimension of var})$$

$$A \in \mathbb{R}^{m \times d} \quad (m: \# \text{ of constraints})$$

feasible $-\infty$

Dual

$$\beta = \max_{y \in \mathbb{R}^m} b^T y$$

$$\alpha = \beta$$

$$\begin{aligned} \text{s.t. } & A^T y \leq c \quad (d \text{ constraints}) \\ & y \geq 0 \quad (m \text{ constraints}) \end{aligned}$$

$$A^T \in \mathbb{R}^{d \times m} \quad \begin{aligned} & (d: \# \text{ constraints}) \\ & (m = \text{dimension of var}) \end{aligned}$$

not feasible $-\infty$

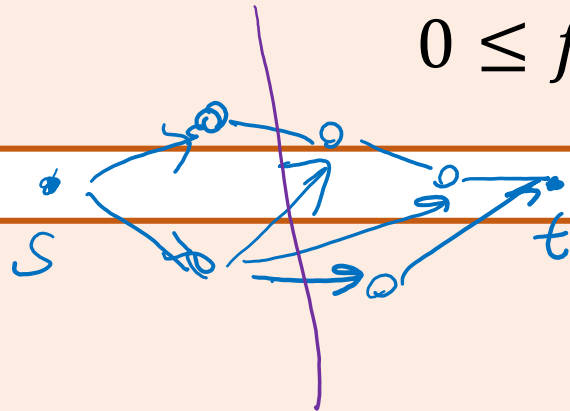
Examples of Primal/Dual Pairs

$$\text{max flow} \leq \text{min cut}$$

Max flow

$$\max_{f, F} F \quad \text{subject to} \quad \sum_{e \in \text{OUT}(v)} f_e - \sum_{e \in \text{IN}(v)} f_e = \begin{cases} F, & v = s \\ -F, & v = t \\ 0, & v \neq s, t \end{cases}$$

$$0 \leq f_e \leq c_e \quad \forall e \in E$$



$\{ S : \text{containing } s \}$
 $\{ T : \text{containing } t \}$

$$\text{value}(\text{cut}) = \sum_{\substack{u \in S \\ v \in T}} c_{uv}$$

min cut

$$\min_{y, z} \sum_{(u, v)} c_{uv} z_{uv} \quad \text{subject to}$$

$$z_{uv} \geq y_u - y_v \quad \forall (u, v) \in E$$

$$z_{uv} \geq 0 \quad \forall (u, v) \in E$$

$$y_s - y_t = 1$$

Examples of Primal/Dual Pairs

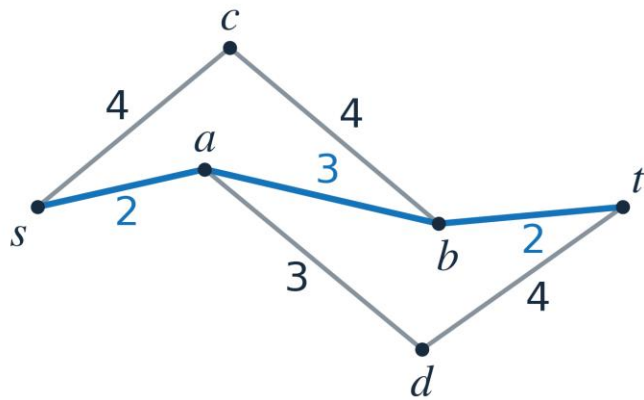
$$\min_{x \geq 0} \sum_{e \in E} c_e x_e \quad \text{subject to} \quad \sum_{e \in \text{OUT}(v)} x_e - \sum_{e \in \text{IN}(v)} x_e = \begin{cases} 1, & v = s \\ -1, & v = t \\ 0, & v \neq s, t \end{cases}$$

$$\max_d d_t - d_s \quad \text{subject to} \quad d_v - d_u \leq c_{uv} \quad \forall (u, v) \in E$$

Examples of Primal/Dual Pairs

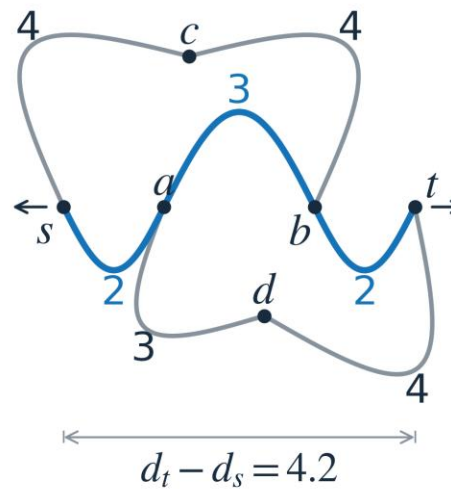
Pull the network apart; the shortest path becomes taut

Original graph

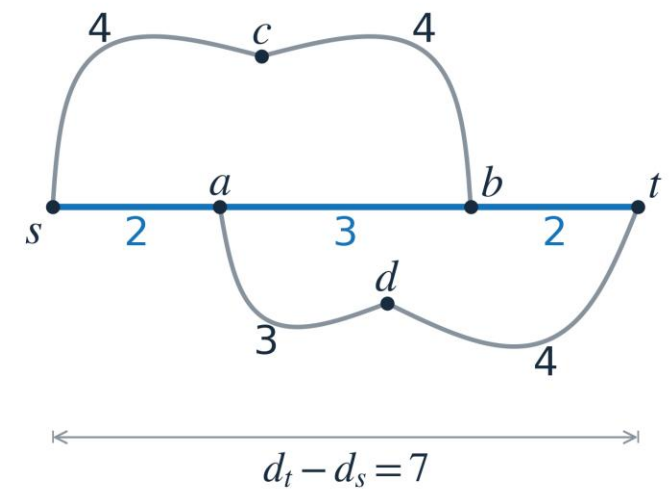


Edge labels = costs

Slack ropes



Maximum separation



Weak / Strong Duality Theorem

Theorem. (Weak Duality Theorem)

$$\overset{\text{min}}{\text{Primal value}} \geq \overset{\text{max}}{\text{Dual value}}$$

Theorem. (Strong Duality Theorem)

If either primal or dual problem is feasible, then

$$\text{Primal value} = \text{Dual value}$$

Weak / Strong Duality Theorem

Primal (min)	Dual (max)	Optimal Value
Feasible	Feasible	$p^* = d^* \in (-\infty, \infty)$
Feasible	Infeasible	$p^* = d^* = -\infty$
Infeasible	Feasible	$p^* = d^* = \infty$
Infeasible	Infeasible	$p^* = \infty, \quad d^* = -\infty$

Proof (Idea) of Weak / Strong Duality Theorems

$$\min_x c^T x$$

$$\text{s.t. } Ax \geq b$$

$$x \geq 0$$

$$\max_y b^T y$$

$$\text{s.t. } A^T y \leq c$$

$$y \geq 0$$

If y violates some constraint

$$(A^T y - c)_i > 0$$

$$\Rightarrow x_i \rightarrow \infty$$

Assume Feasible Primal

$$\min_{x \geq 0} c^T x \quad \text{s.t. } Ax \geq b$$

$$= \min_{x \geq 0} \max_{y \geq 0} c^T x - y^T (Ax - b)$$

$$\frac{DGJ^T}{T}$$

Dual

$$\max_{y \geq 0} b^T y \quad \text{s.t. } A^T y \leq c$$

$$= \max_{y \geq 0} \min_{x \geq 0} b^T y - \underbrace{x^T (A^T y - c)}_{\substack{\uparrow \\ +\infty \quad > 0}}$$

Proof (Idea) of Weak / Strong Duality Theorems

Solving LP with No-Regret Algorithms

Arora, Hazan, Kale. The Multiplicative Weights Update Method: A Meta-Algorithm and Applications. 2012.

(Problem 2, Homework 2)

Linear Programming

$$\min_x c^\top x$$

$$\text{s.t. } Ax \geq b \quad (\text{difficult constraint})$$

$$x \in \mathcal{K} \quad (\text{easy constraints})$$

$$= \min_{x \in \mathcal{K}} \max_{y \geq 0} c^\top x + y^\top (b - Ax)$$

Assumption: solving $\min_{x \in \mathcal{K}} g^\top x$ is easy

If there is no obvious easy constraints $\mathcal{K}$, we can always choose $\mathcal{K} = [-B, B]^d$

$$\min_{x \in [-B, B]^d} g^\top x \quad \text{is easy}$$



$$\begin{cases} g_i \geq 0 & \Rightarrow x_i = -B \\ g_i < 0 & \Rightarrow x_i = B \end{cases}$$

Linear Programming

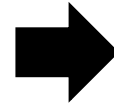
infeasible U^* *feasible*

Find $x \in \mathcal{K}$ such that

$$c^T x \leq \mathbf{U}$$

$$Ax \geq b$$

or certify no such x exists



Find $x \in \mathcal{K}$ such that

$$\tilde{A}x \geq \tilde{b}$$

or certify no such x exists



If the problem is feasible

$$\min_{x \in \mathcal{K}} \max_{y \in \Delta_N} y^T (\tilde{b} - \tilde{A}x) \leq 0$$

If  *infeasible*

$$\min_{x \in \mathcal{K}} \max_{y \in \Delta_N} y^T (\tilde{b} - \tilde{A}x) > 0$$

$$\min_{x \in \mathcal{K}} \max_{y \in \Delta_N} y^T (\tilde{b} - \tilde{A}x)$$

$$\operatorname{argmin}_{x \in \mathcal{K}} \max_{y \in \Delta_N} y^T (\tilde{b} - \tilde{A}x)$$

Using No-regret algorithm to solve LP

Initialize a no-regret algorithm **ALG** over Δ_N

For $t = 1, \dots, T$:

Let **ALG** output $y_t \in \Delta_N$

Find $x_t = \operatorname{argmin}_{x \in \mathcal{K}} y_t^\top (\tilde{b} - \tilde{A}x)$

Define **reward** $r_t = \tilde{b} - \tilde{A}x_t$.

Feed reward vector r_t to **ALG**.

$$\bar{x} = \frac{1}{T} \sum_{t=1}^T x_t \in \mathcal{K}$$

$$\min_{x \in \mathcal{K}} \max_{y \in \Delta_N} y^\top (\tilde{b} - \tilde{A}x) \leq \max_{y \in \Delta_N} y^\top (\tilde{b} - \tilde{A}\bar{x}) \leq \min_{x \in \mathcal{K}} \max_{y \in \Delta_N} y^\top (\tilde{b} - \tilde{A}x) + \sqrt{\frac{\log N}{T}}$$

Need another outer loop of binary search over **U**

Methods for Solving Linear Programming

Method	Time	Property
Interior point	$\text{Poly}(m, n, \log(1/\epsilon))$	General, high accuracy
Simplex	Exponential in n in the worst case	Often fast, exact solution
Ellipsoid	$\text{Poly}(m, n, \log(1/\epsilon))$	Mainly theoretical
No-regret	$\text{Poly}(\mathbf{m}', n, 1/\epsilon, \text{time of best-response oracle})$	Cheap best-response oracle Looser accuracy requirement

m : # constraints

n : # variables

m' : # difficult constraints